UNIVERSITY of RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

DEPARTMENT OF COMPUTER ENGINEERING

Mobile Application Systems and Design

Push Notifications with Firebase Cloud Messaging (FCM)

Registration number:

223008047

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Year 3

Lecturer: Ukulikiyeyesu (Mr Dieudonne)

Done at 8th April 2026

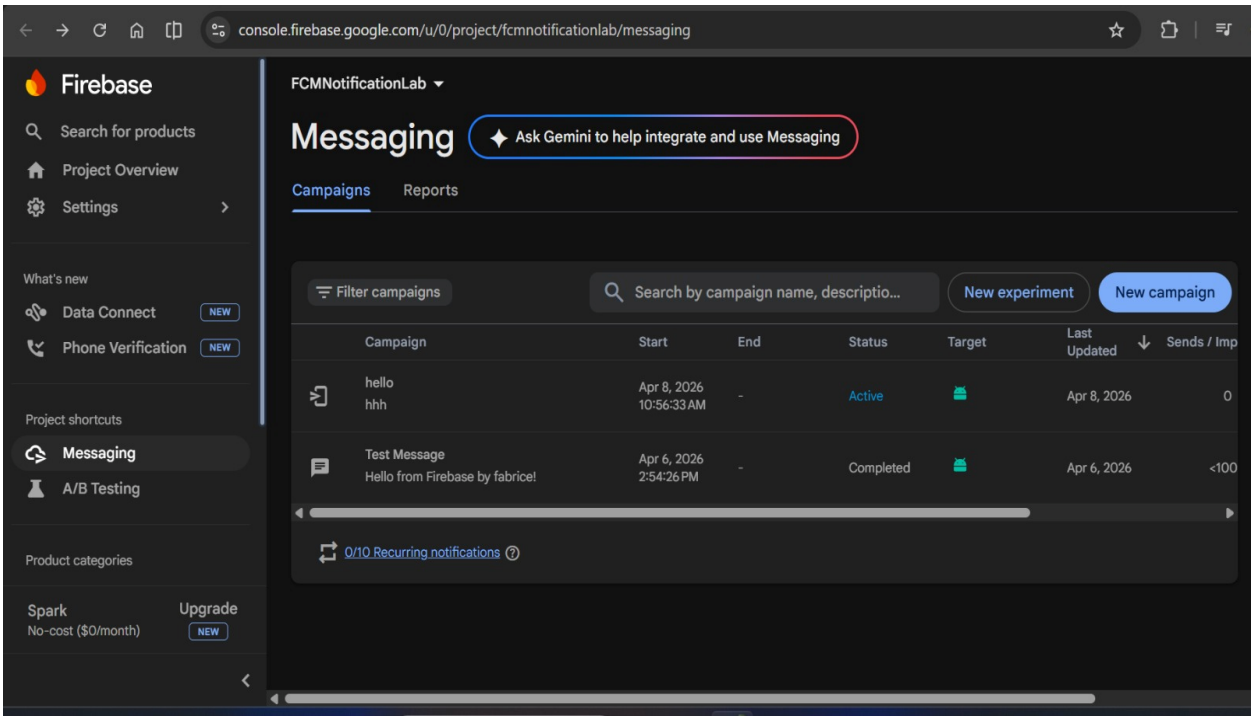

Fig1 :Firebase cosole messaging

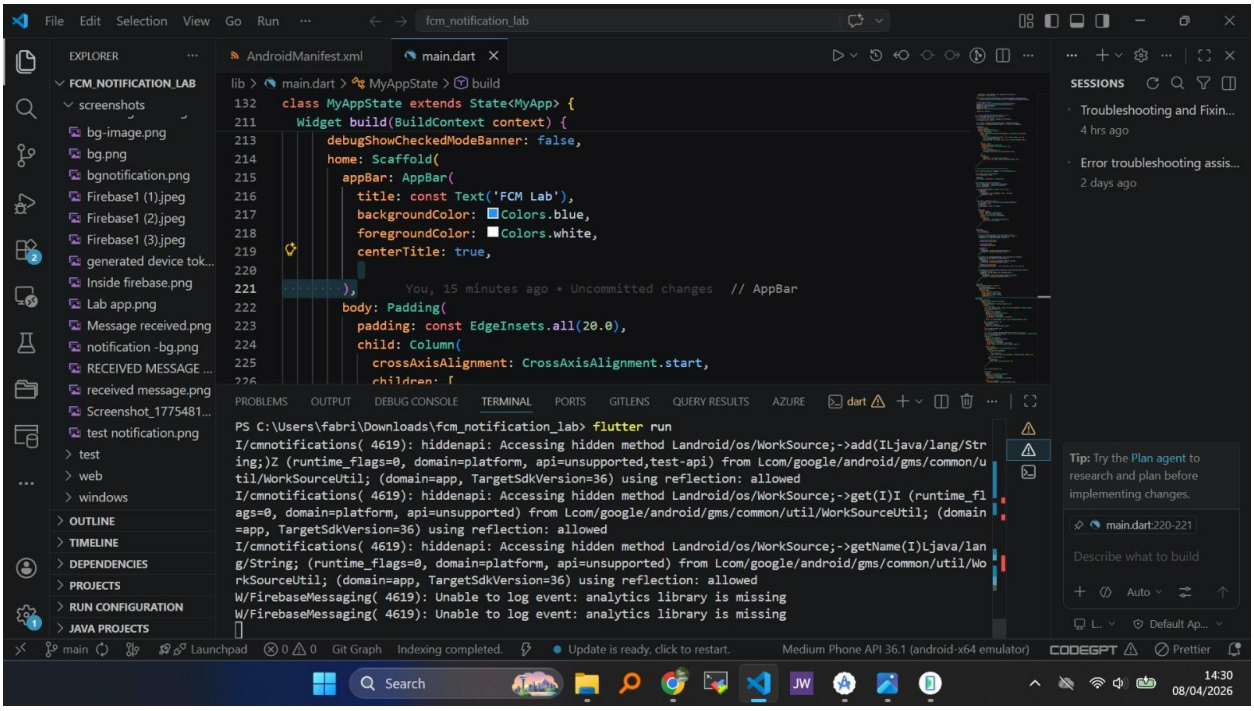

Fig 2:Inside VS code

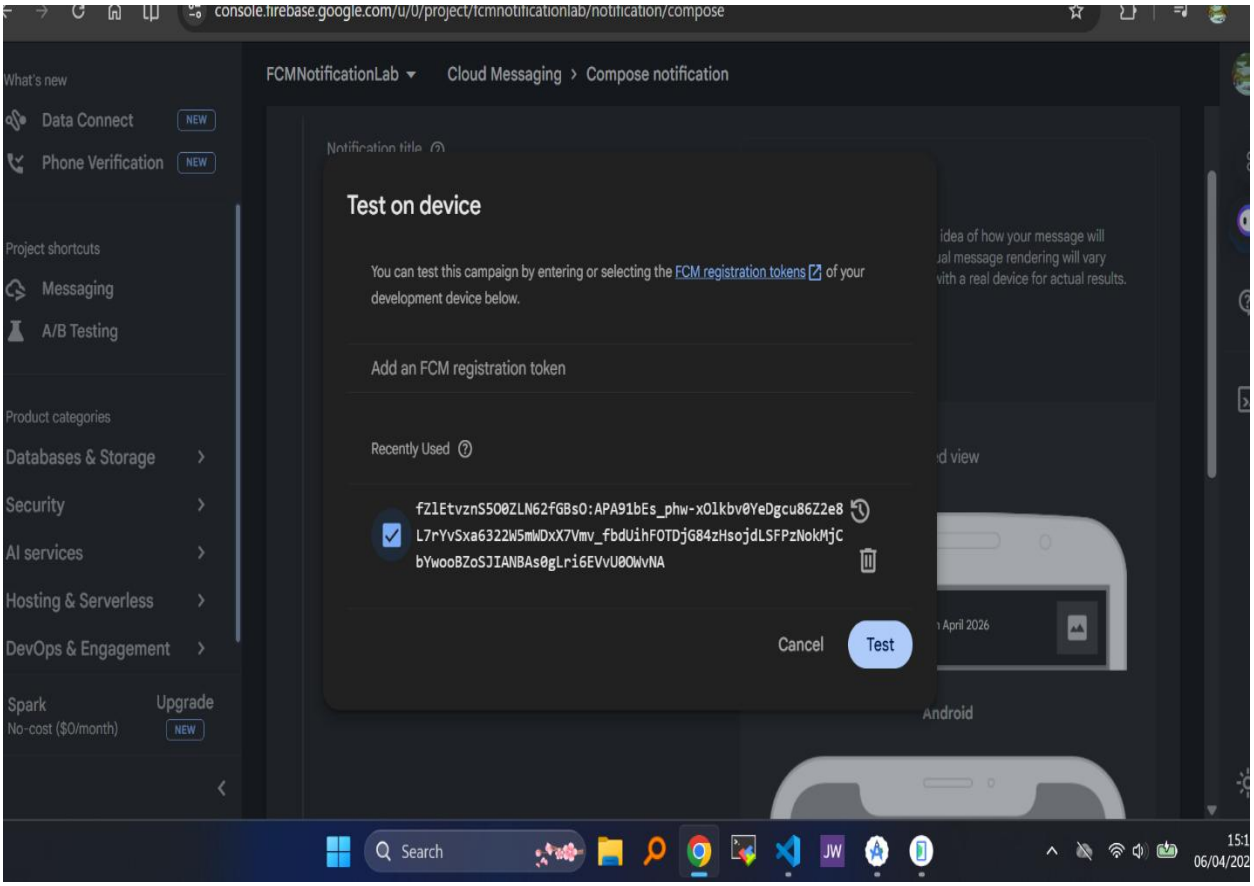

Fig 3:Test using generated Token

Fig 4:background notification

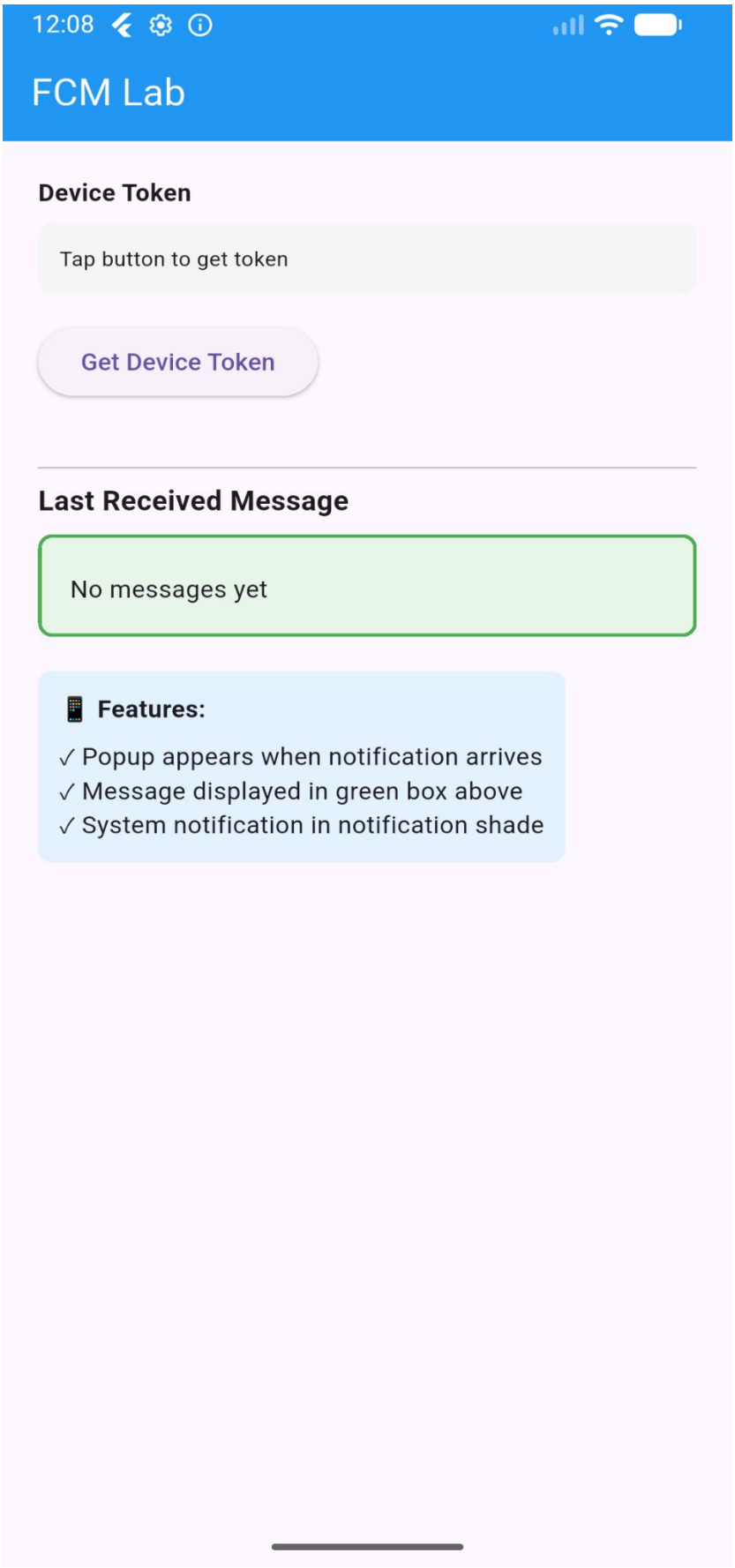
12:08
FCM Lab
Device Token
Tap button to get token
Get Device Token
Last Received Message
No messages yet
Features:
✓ Popup appears when notification arrives
✓ Message displayed in green box above
✓ System notification in notification shade

Fig 5:FCM Lab

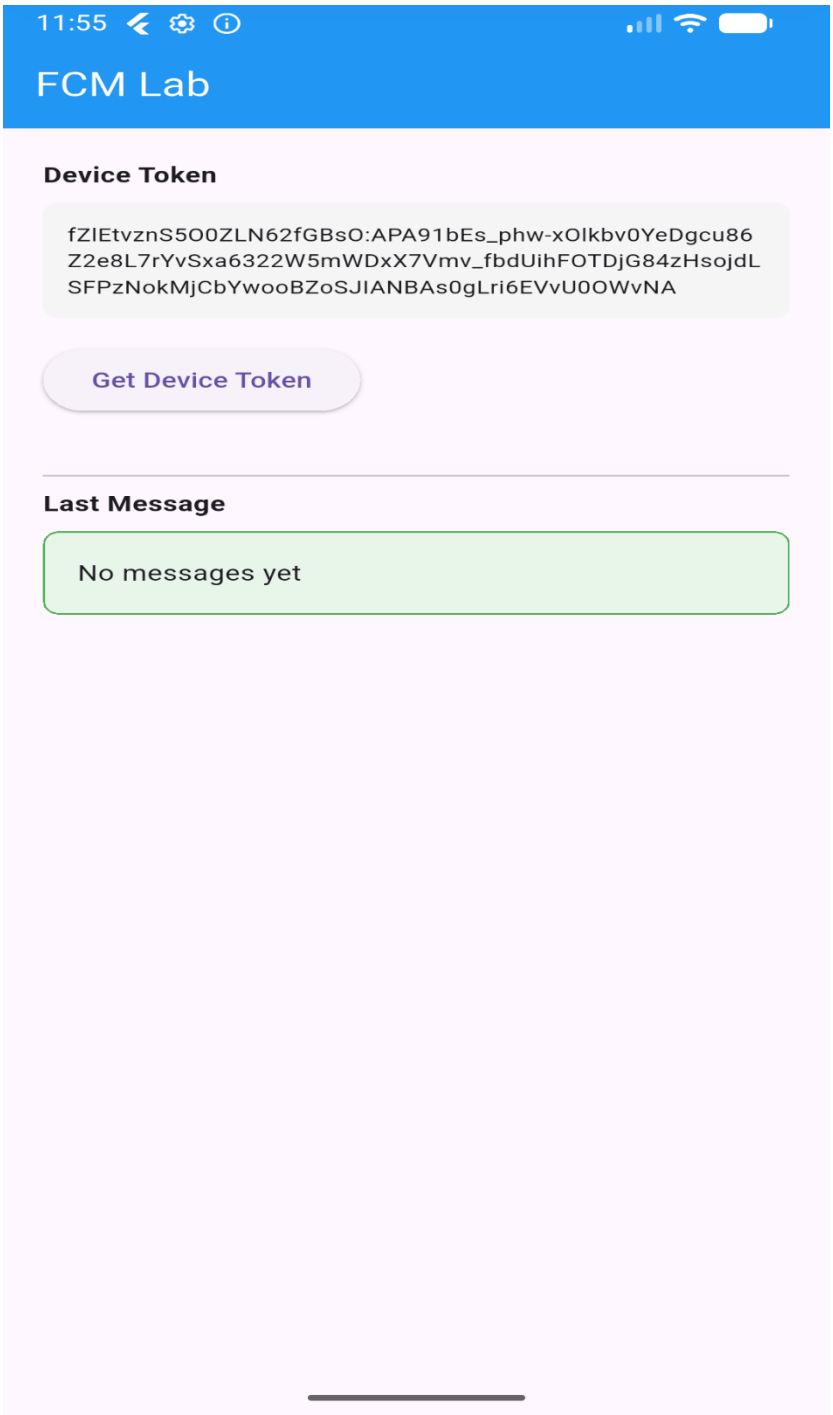

Fig 6:Generated token

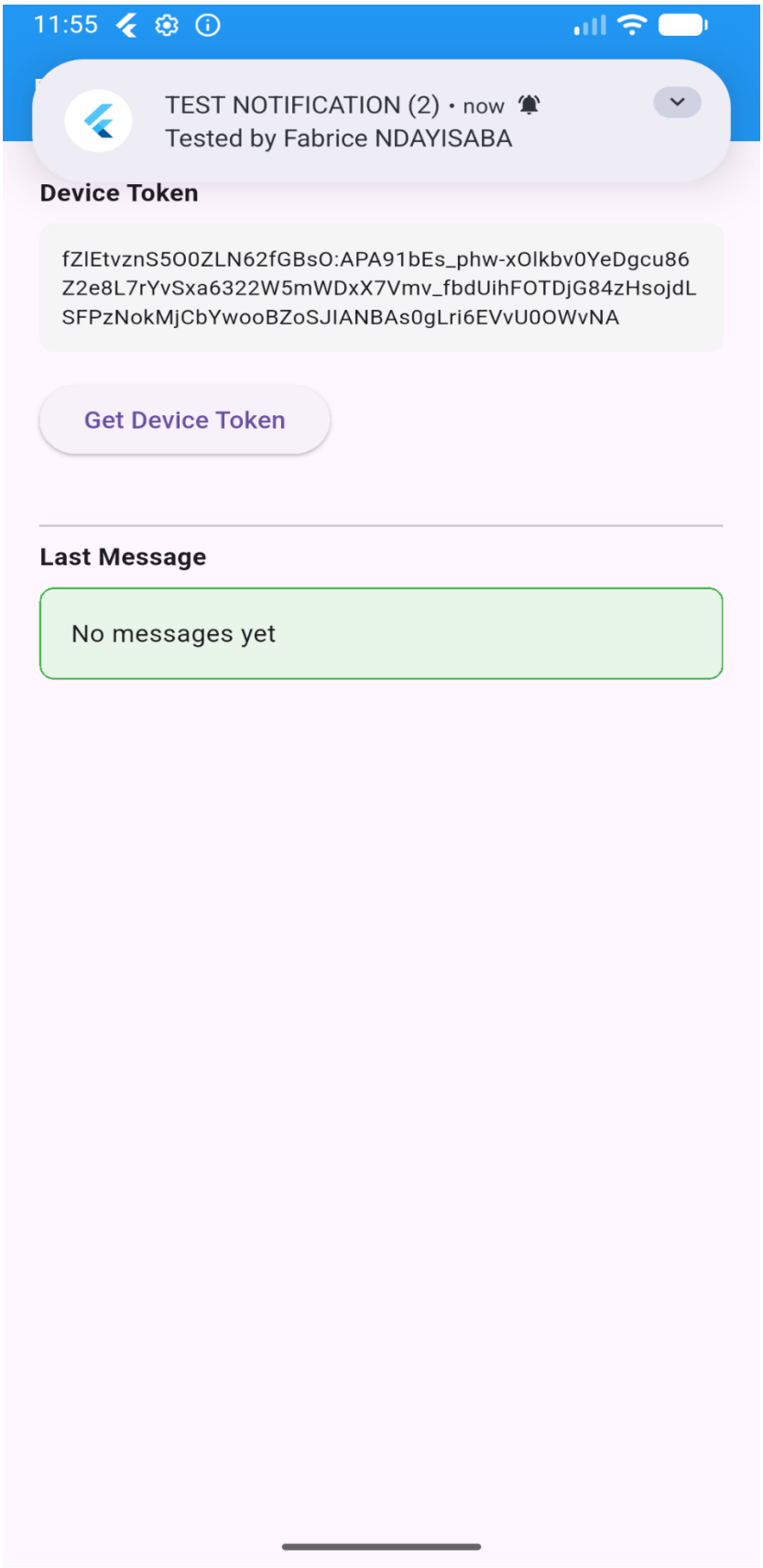

Fig7:Notification pop up

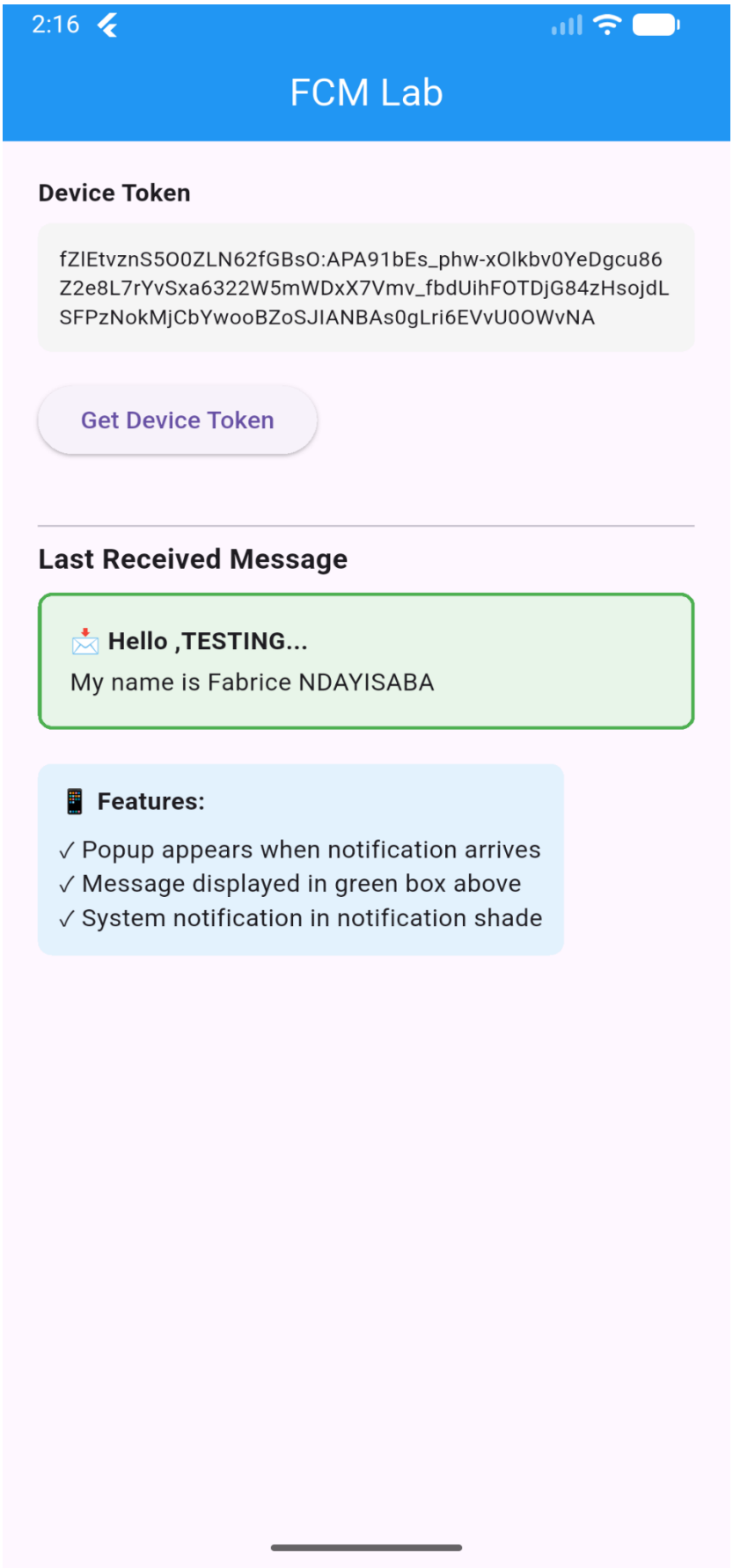

Fig 8: Last Received message

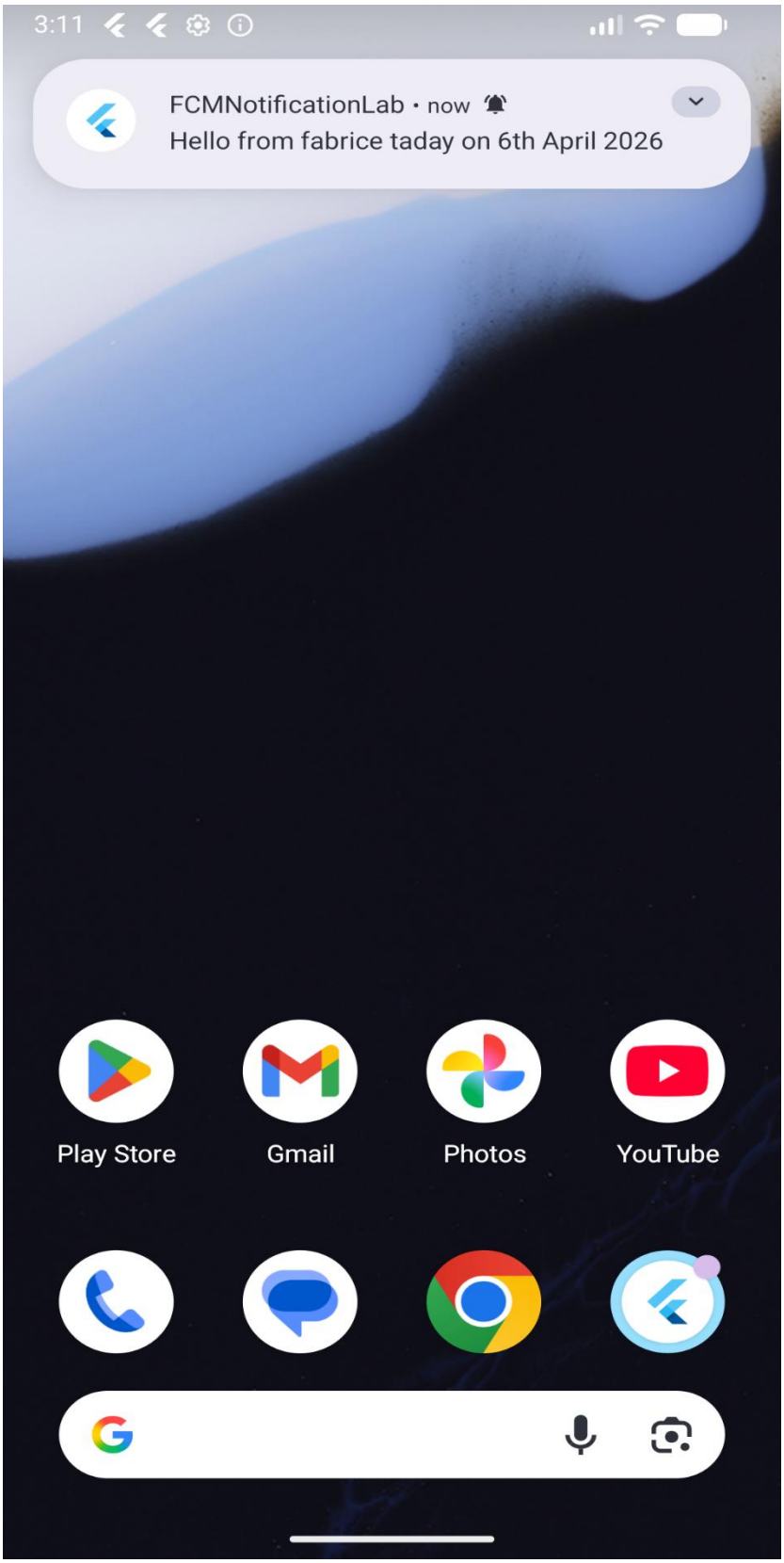

Fig 9:Background notifications.

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Module Name: Mobile Application systems and design.

Reg No: 223008047

on 8th April 2026.

My observations

1. Foreground Vs Background Notifications: When the app is open, notifications appear as popups immediately. When the app is in the background, notification appear in the system notification tray.

2. Device Token uniqueness: Each device gets a unique FCM registration ~~Send~~ token. This token is required to send notification to specific devices.

3. permission requirement: Android 13+ requires explicit user permission before showing notification. the app must request "POST_NOTIFICATION" permission.

4. FCM message Types: Firebase supports notification messages (auto-displayed) and data messages (handled by app).

5. Physical device; foreground Vs background:

Foregrounds: - pop up notification appears immediately at the top of the screen.
- sound / Vibration plays according to device notification settings.
- Console logs shows "message received!".

Backgrounds: - Notification appears in system notification shade.
- Sound / Vibration plays.
- App icon badge appear.

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Key steps to follows:

① Enabled 2SV (MFA) on google account to access Firebase Console

② Created Firebase project named "FCMNotificationLab" with package name : com.fabric.fcmnotifications.

③ Downloaded and placed google-services.json in android/app/ directory

④ Added dependencies to pubspec.yaml:

- firebase_core
- firebase_messaging
- flutter_local_notifications.

⑤ Android file configuration

=> android/build.gradle.kts

Added google services classpath.

=> android/app/build.gradle.kts.

Added Firebase BOM and FCM dependencies.

Enabled core library desugaring for flutter_local_notifications.

=> AndroidManifest.xml

Added FCM channel meta-data inside <application> tag. It tells Firebase which notification channel to use for displaying popup notifications on physical device.

⑥ Flutter implementation

Notification permission: Added code to request notification permission on app Launch sections (alert, badge, sound: True);

ON device; system dialog appears asking "Allow FCM to send notifications?"

— tap "Allow".

Retrieved displayed device token using FirebaseMessaging.instance.

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Message listeners:

- set up FirebaseMessaging.onMessage.listen() for foreground messages.
- set up FirebaseMessaging.onMessageOpenedApp.listen() for background messages.

Popup Notification Functionality: created high-importance notification channel for popup display

⑦ Testing on device:

* Connecting Android phone via USB with USB debugging enabled.

⇒ Ran flutter clean, flutter pub get, flutter run.

⇒ On First Launch: Granted notification permission when system dialog appeared.

Message Delivery:

- sent test notification from firebase Console using device token.
- Message successfully displayed in green box on app UI.
- Console logs Confirmed: "message received!"

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